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India
SMART UTILITY
Week 2026
10 – 14 March 2026, New Delhi

Session : NEW TECHNOLOGIES: AI–ML–VR–AR AND ROBOTICS APPLICATIONS AND RELIABLE COMMUNICATIONS FOR UTILITIES

STRATEGIC BLUEPRINT FOR AN INTELLIGENT AUTONOMOUS GRID

Presented By

Meenakshi Vashist – Founder & CEO TekUncorked

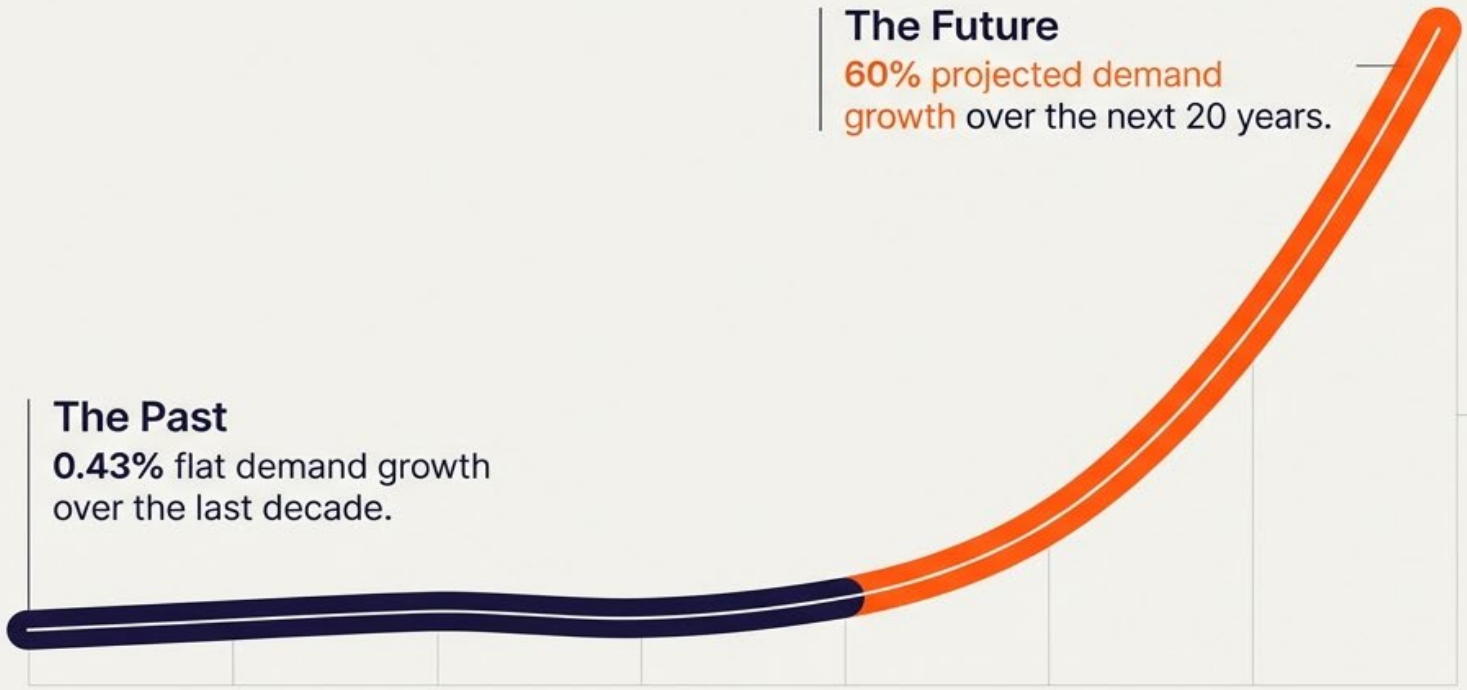
TekUncorked

The 2026 Utility Landscape

A Strategic Blueprint for an Intelligent, Autonomous Grid

TekUncorked

The Era of the Flat Load Curve is Permanently Over

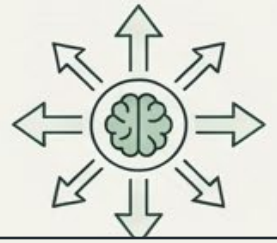


The Driver

Data center campuses demanding **300–500 MW** of power, with 1 GW hyperscale facilities built in months, not years.

Explosive demand from AI data centers, reshoring, and the electrification of everything forces a total departure from incremental infrastructure planning.

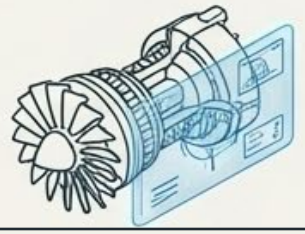
Four pillars define the AI-enabled utility.



Grid Resiliency

Autonomous power flow, renewable integration, and outage simulation.

TekUncorked's GridIoT.ai platform has successfully demonstrated Grid Monitoring -> Analysis -> Control flow to enable self healing grids



Asset Optimization

Predictive maintenance and virtual replicas.

TekUncorked's GridIoT.ai platform provides Asset Health Indexes, Fault Vulnerability, EoL Forecasting and Location-enabled 'Live' replica of Substation and Network SLDs

Workforce & Training

AI virtual mentors and digital twin environments.

TekUncorked's GridIoT.ai platform has successfully demonstrated Grid Monitoring -> Analysis -> Control flow to enable self healing grids

Customer & Compliance

Dynamic forecasting and automated regulatory reporting.

Reduced Outages, Faster Recovery & Enhanced power quality due to AI led digitization of the grid increases customer satisfaction by more than 80%

Scaling Grid Efficiency with a Unified Layered Architecture

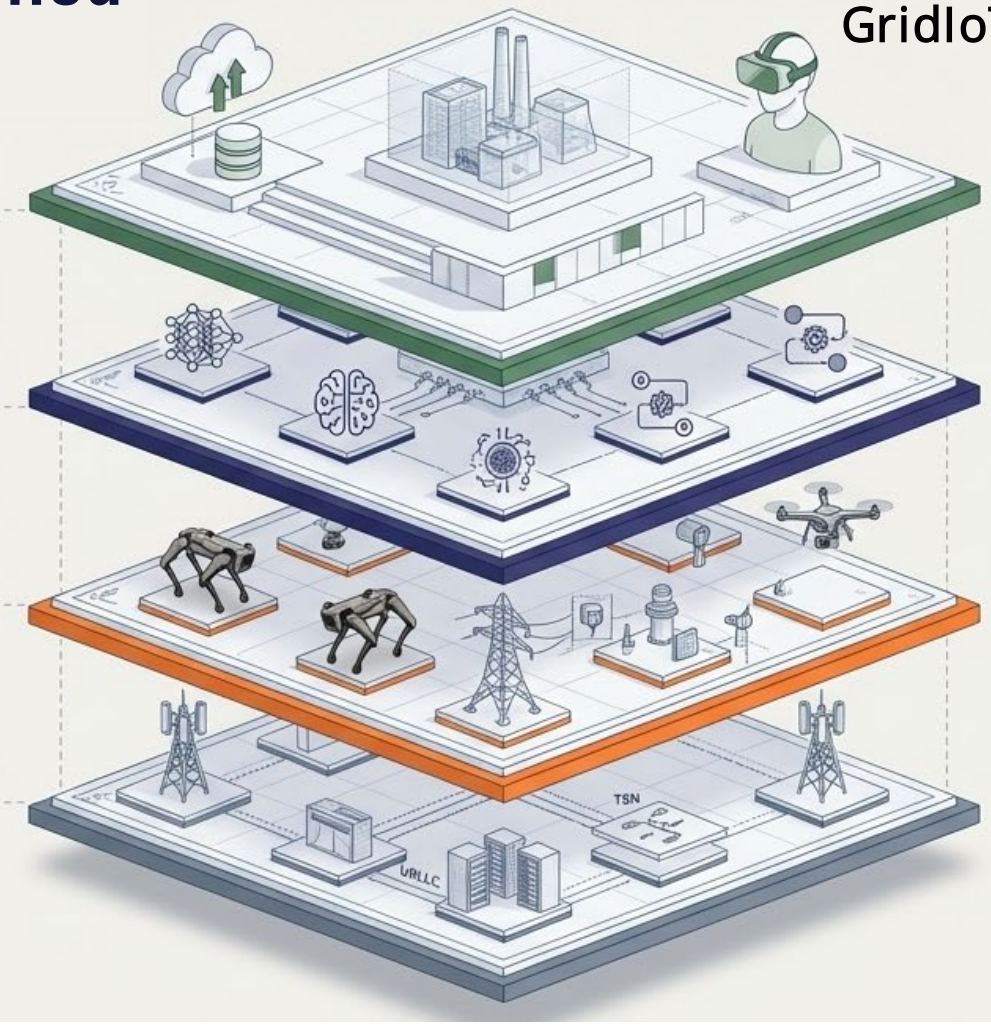
GridIoT.ai

Layer 4: Modern Operations
New Age Applications like Digital Twin, AI-led OMS, Cloud FinOps

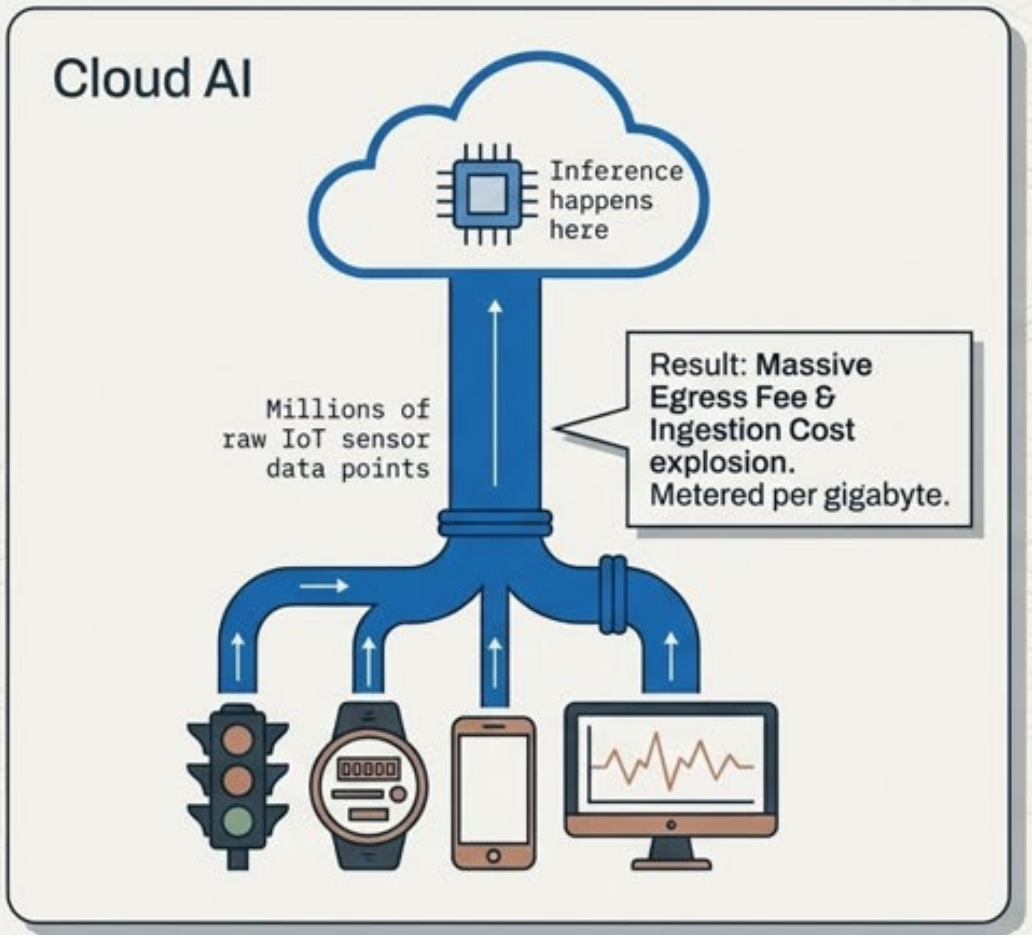
Layer 3: Intelligence Engine
Contextual Correlations, Predictive AI, Machine Learning, Agentic Workflows

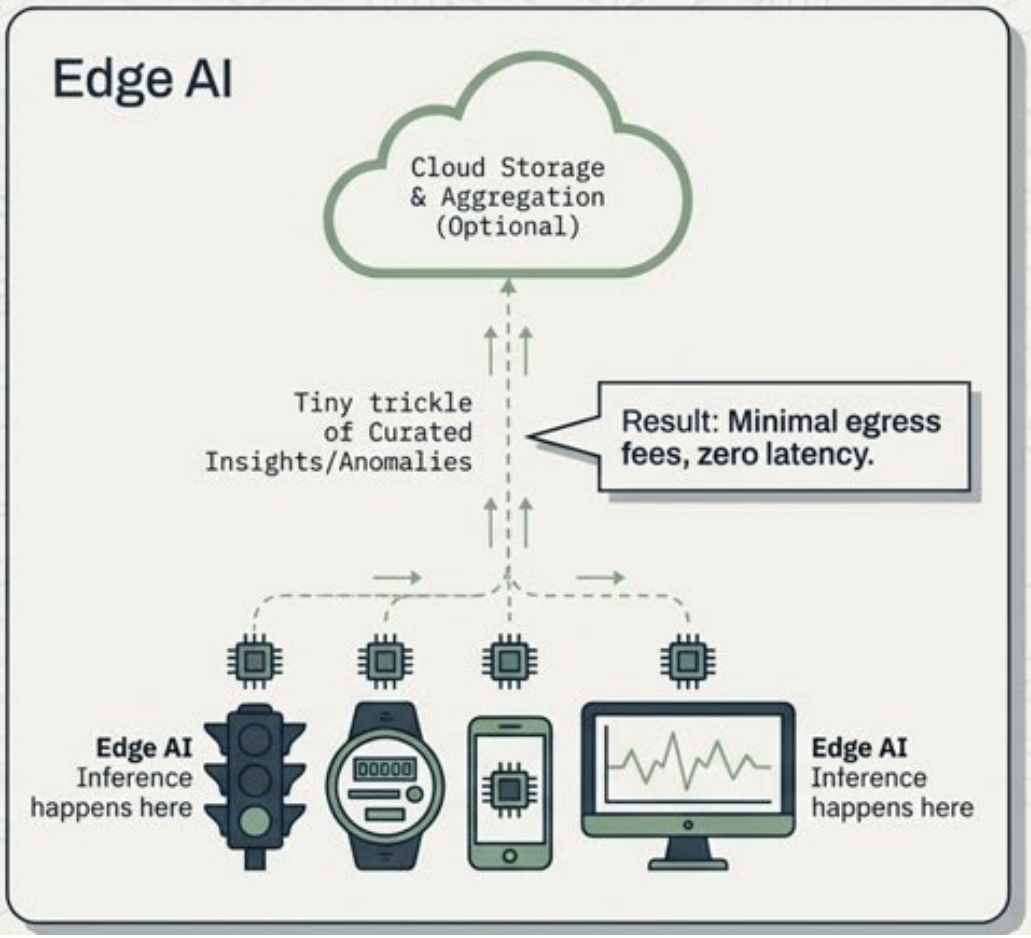
Layer 2: Autonomous Edge
Smart Sensors, Data Adapters, Control Actuators

Layer 1: Connectivity Backbone
Secure 5G M2M Backhaul



Hybrid AI Architecture chosen to prevent economic data egress overload which may subsequently put a pressure on consumer tariff



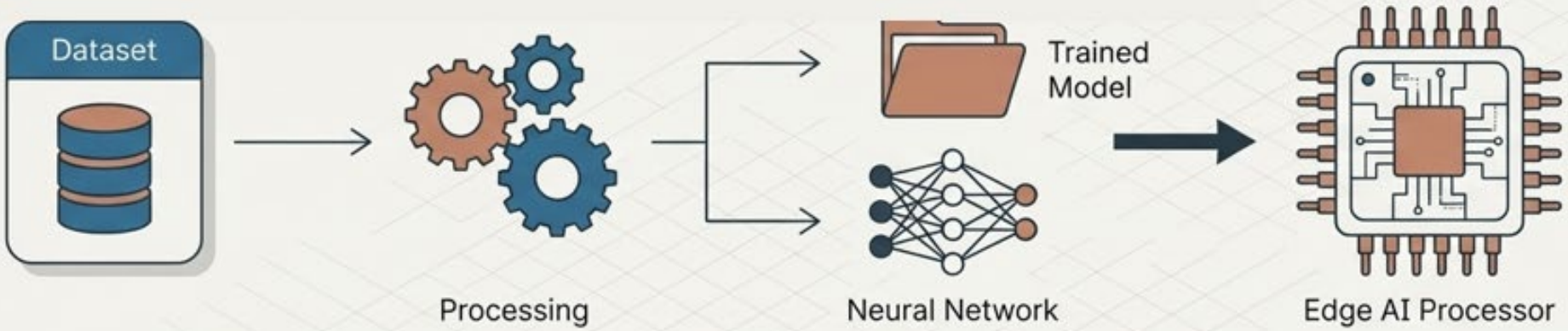


Hybrid AI Architecture sets an adaptive framework for Micro as well as Macro Grid Management

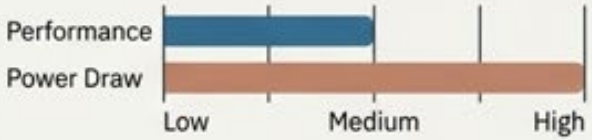
Architecture	Latency	Scalability	Energy Efficiency	Privacy	Primary Utility Use Case
Cloud AI	High (Network Dependent)	Infinite	Low (Costly transport)	Vulnerable	Demand forecasting, complex historical modeling
Edge AI	Zero (Real-time)	Hardware Constrained	High	Secure (Local)	Autonomous substation security, valve control
Hybrid AI	Adaptive	High	Optimized	Tiered	Enterprise-wide smart grid management

Fast Forward: Edge AI through Edge SLMs powered by Neural Processing Units

Context: Small Language Models (SLMs) bring reasoning to resource-constrained edge devices without the bloat of LLMs.

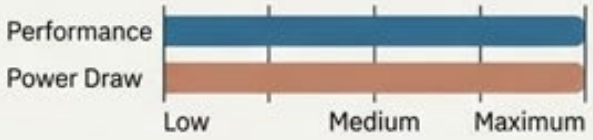


CPU



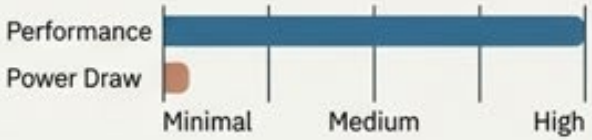
General purpose. Struggles with parallel AI tasks. High energy drain.

GPU



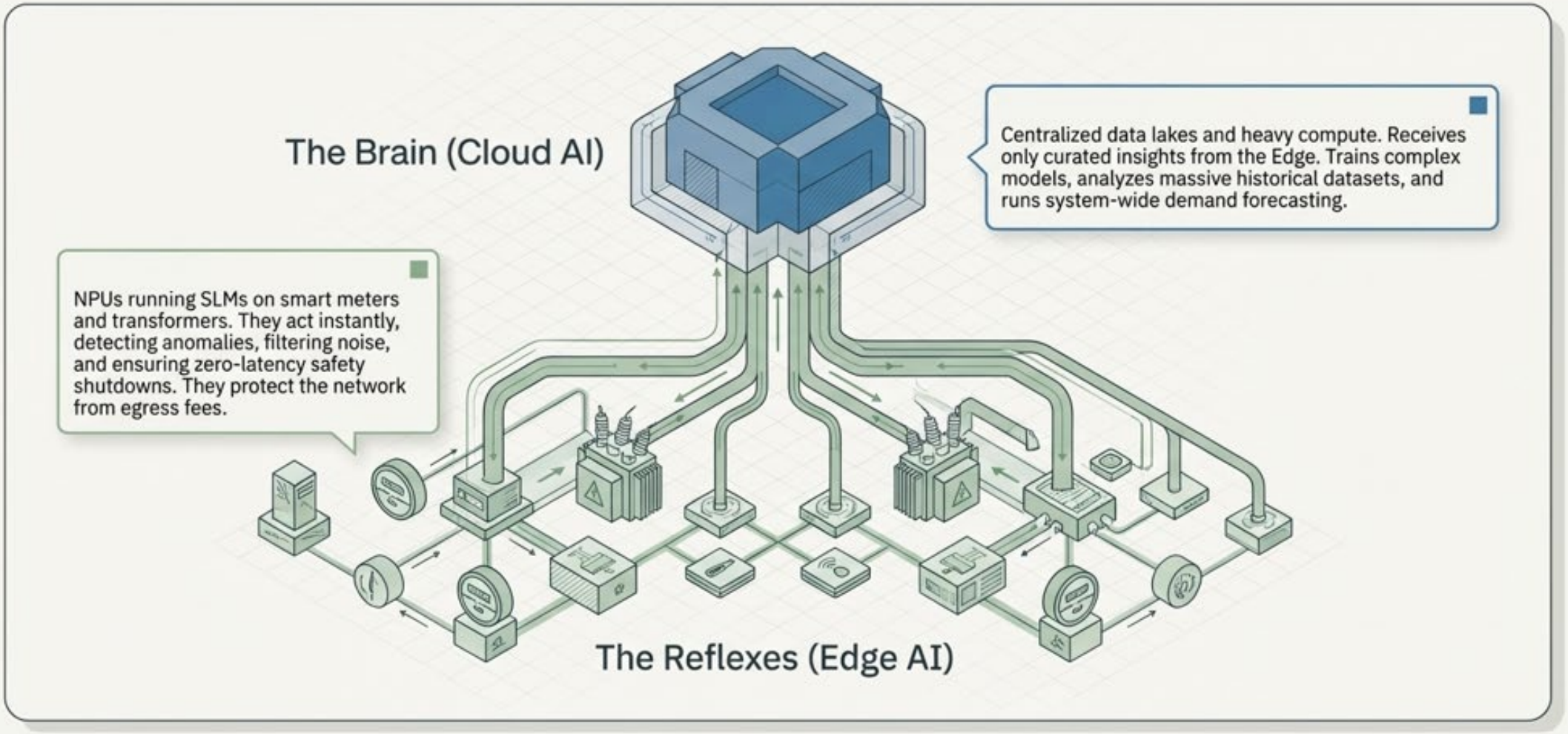
Massive performance, but highly power-intensive. Not viable for remote IoT battery-operated sensors.

NPU (Winner)



Custom-designed for AI tensor math. Delivers the highest performance per watt (EDP). The mandatory standard for remote utility edge deployment.

Hybrid Ecosystem for economically feasible and scalable AI Architectures in Utilities



The Blueprint Directives

01 Mandate the Hybrid Model

Do not fall into the Cloud-only trap. Designate the Edge for real-time reflexes and the Cloud for deep historical analysis.

02 Control the Edge to Control Costs

Deploy SLMs on Neural Processing Units (NPUs) to filter IoT data locally, ruthlessly cutting cloud egress fees and latency.

03 Fix the Data Pipeline First

AI is only as good as the data feeding it. Invest in integration, legacy system modernization, and secure vector databases before buying expensive models.

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